

Table 5
Clusters of perceived usefulness — AI literacy.

Question	C_1 (mean)	C_2 (mean)	Q(Average)
Q1	3.40	3.0	3.20
Q2	3.33	3.25	3.29
Q3	3.96	2.91	3.70
Q4	4.11	3.25	3.435
Q5	4.55	2.75	3.65
TotalCount	27	12	
Average	3.87	3.03	

Table 6
Perceived usefulness and associated AI literacy, survey questions.

Question	Perceived usefulness
Q1	Can you indicate your level of knowledge on diagnostic medicine
Q2	Computer vision is a field of artificial intelligence that trains computers to interpret and understand the visual world. Would you trust it as a feature in driving automation
Q3	AI can add value by either automating, assisting or augmenting the work of clinicians and staff. Many repetitive tasks will become fully automated, and we can also use AI as a tool to help health professionals perform better at their jobs and improve outcomes for patients. How much do you agree with the above statement
Q4	If using AI-empowered systems do you feel that the more you know about the underlying technology the better, or you had trust the provider
Q5	Would you participate in developing AI medical applications?

reinforcing our approach to employ AI literacy as a metric for this external variable. Table 6 illustrates the selected questions that estimate perceived usefulness and corresponding AI literacy. Q1 gauges the participants' understanding of diagnostic medicine, a domain that employs statistical methods and data analysis to enhance diagnosis and treatment decisions. The participants' grasp of diagnostic medicine suggests a comprehension of statistics and data analysis, thus hinting at an elevated AI literacy level. Q2 aims at revealing the degree of understanding and trust in computer vision, particularly when employed in autonomous driving, a widely commercialised application of this technology. The assumption is that increased trust corresponds to a more profound understanding and a higher AI literacy level in respondents. Similarly, Q3 and Q4 outline fundamental functionalities of AI, thereby reflecting the respondents' AI comprehension and literacy. Finally Q5, queries on the willingness of the participants to take part in developing AI-empowered medical systems. The scale of the response indicates a higher perception of usefulness, with regard to this particular technology.

Employing the same approach, hierarchical clustering is utilised. As per the silhouette coefficients, two clusters are identified as optimal (see Fig. 11). These clusters can be observed in Fig. 11, where Principal Component Analysis (PCA) has been employed to reduce the dimensions of the space of features-questions from 5 to 3. Question Q2, which assesses the trust in computer vision as a driver for automated driving, scores the lowest. In contrast, question Q5 scores the highest, illustrating an elevated inclination for involvement in the evolution of AI-empowered medical systems.

Using the score system proposed in Table 5, we identify 2 levels of AI literacy.

- Expert AI literacy level, clusters: C1
- Average AI literacy level, clusters: C2

By utilising the rule-based system described previously, the implementation strategy, which hinges on the clustering delineated by the two levels of PINXEL, will proceed as outlined:

- (C1) IF EAIL(3) and MLA(2) THEN Explainability Depth is 6 USE Corresponding levels
- (C2) IF AAIL(2) and MLA(2) THEN Explainability Depth is 4 USE Corresponding levels

In summary, the subjects outlined in Figs. 4 and 5, are to be explained, using a Medium level of abstraction, i.e more technical questions to be answered by NLPs empowered by Large language models (LLMs), as proposed in Fig. 6.

5. Implementation examples in NuhealthSoft

In Fig. 12, an outline of the layers of the system are presented. The first layer contains the database (D.b) controllers, which includes the business logic of the system and the ML algorithms that provide classification and recommendation methods. The second layer contains the structural semantics, i.e. the inputs, outputs, stakeholders and associated context, which are extracted by the D.b controllers and provide the semantic groupings as XAI objects. Also in this second layer, the json summaries of the different ML parameters are stored, again as objects to be utilised in user interfaces where different AI literacy levels and levels of abstraction are taken into account, as outlined in the previous sections. These user interfaces are defined in the final (third) layer of the system, which contains the proposed transparency and XAI framework, as described previously, and implemented in accordance with the pre-set explainability depth (Expert, average, basic etc.).

5.1. Addressing concerns, towards building responsible systems

All the different parameters discussed in this paper, encompass what can generally be perceived as a strategy towards transparency, interpretability and inclusiveness. In the following subsections, some different use cases are provided in accordance to PINXEL, and implemented in NuhealthSoft.

5.1.1. PINXEL — average AI literacy level with low level of abstraction

In Fig. 13, an implementation example can be seen for an average literacy level with Low level of abstraction. The architecture part of this figure demonstrates the different layers of the trained model. It consists of an input layer of 18 neurons in total. The input layer receives the raw data in the form of feature vectors. The output is a transformed version of the input layer which is passed to next layer, if one exists. The second layer, named “dropout,” is a regularisation technique that prevents over-fitting by randomly dropping out neurons during training. In this case, 50% of the neurons from the previous layer are dropped and fed to next layer. The last dense layer extracts a binary prediction or classification. The parameters tab refers to the training parameters of the neural network, i.e. the weights and the biases. These parameters break down the symmetry of the fed input to ensure that different neurons learn different features. Then the weighted sum of inputs of each neuron is calculated using an activation function.

Using optimisation techniques, loss calculation and continuously updating the parameters, the weights and biases change towards the optimum result, i.e. towards a prediction or classification both on training and testing data. In this case the parameters change for each layer, since a dropout layer is being applied. The formula to calculate the total parameters for each layer can be seen below: parameters = (input neurons + 1) * output neurons. The +1 accounts for the bias associated with each output neuron.

In the same figure, we can see the distribution of the weights and biases of the first layer. The confusion matrix shows the relevant accuracy of the final trained model on the testing data. Finally, the history plots show how the training-validation loss and accuracy change over time, in the case of training a neural network over a series of training epochs.